

CS661: Project Proposal Report

GROUP-2

1 Introduction

Cricket is one of the most popular sports globally, generating extensive data from matches played across different formats (Test, ODI, and T20). This data includes detailed ball-by-ball records, player statistics, and match outcomes, which, when analyzed, can provide valuable insights into team performances, player efficiencies, and match conditions.

Our project aims to develop an interactive web-based data visualization system that provides insights into cricket match analytics. Using the Cricsheet dataset (<https://cricsheet.org/matches/>), we will analyze and visualize various statistics such as average innings scores per ground, player performances under different conditions, and team-based trends over time.

The system will offer interactive charts and dashboards to help cricket enthusiasts, analysts, and coaches gain meaningful insights from historical match data.

2 Data Source

The primary data source for our project is the Cricsheet dataset, which provides ball-by-ball data for various cricket matches in JSON format. This dataset contains:

- Match details (venue, teams, date, format)
- Player performances (runs scored, wickets taken, strike rates, economy rates)
- Ball-by-ball events (deliveries, extras, dismissals)
- Tournament and series information

3 Specific Tasks

For this project, we will perform the following main tasks:

- **Data Aggregation:** Extract and compile match data from Cricsheet into a structured format, ensuring compatibility for analysis.
- **Data Preprocessing:** Clean and process the dataset by handling missing values, removing inconsistencies, and transforming data for visualization.
- **Data Visualization:** Develop interactive visual representations of cricket statistics, trends, and player performances.
- **Insights Generation:** Implement statistical and analytical methods to uncover hidden patterns in the data, such as team performance trends and player efficiency metrics.
- **User Interaction:** Integrate interactive features allowing users to filter, compare, and query specific data points to gain deeper insights.

4 Visualization Tasks

We will implement the following key visual analytics tasks to extract meaningful insights from the dataset:

4.1 Average Innings Score per Ground

Understanding how different stadiums influence scoring patterns is crucial in cricket. This visualization will allow users to:

- Filter by country, stadium, and innings to view the average score for the selected conditions.
- Compare first and second innings average scores across different grounds.
- Identify grounds that favor high-scoring matches or low-scoring encounters.

Suggested Visualization: Interactive bar charts or heatmaps showing average scores, with filtering options for stadium, country, and innings.

4.2 Performance of Individual Players vs Specific Teams

This visualization helps analyze how specific players (batsmen or bowlers) perform against different teams in various formats and tournaments. Users can:

- Select a batsman or bowler and view their runs scored or wickets taken against a specific team.
- Filter by match format (Test, ODI, T20) and tournament.
- Identify player strengths and weaknesses against particular opponents.

Suggested Visualization: Line graphs or scatter plots comparing performance metrics across different teams, with filtering for player, team, format, and tournament.

4.3 Team-Based Performance: Head-to-Head Comparison

A key aspect of cricket analysis is understanding how teams perform against each other. This visualization will:

- Compare win/loss records between two teams.
- Display average scores, best individual performances, and match statistics.
- Allow filtering by tournament and time period.

Suggested Visualization: Side-by-side bar charts or line graphs comparing team statistics over different matches and tournaments.

4.4 Tournament-Based Analysis

This visualization will provide insights into how teams have performed across different tournaments, tracking trends in runs scored, wickets taken, and match outcomes. Users can:

- Select a tournament and analyze team performance over multiple matches.
- View scoring trends, top performers, and match results.
- Compare team consistency across different tournaments.

Suggested Visualization: Tournament-wise stacked bar charts or timeline graphs showcasing trends in runs, wickets, and match outcomes.

4.5 Dismissal Type Analysis

Understanding how players get dismissed helps in strategy formulation. This visualization will:

- Display a breakdown of dismissal types (bowled, caught, LBW, etc.).
- Allow filtering by format, tournament, and player.
- Compare dismissal patterns of top batsmen.

Suggested Visualization: Pie charts or bar graphs with interactive filtering options for player, format, and tournament.

5 Overall Solution

Our proposed solution will be a web-based interactive dashboard that allows users to explore cricket statistics through various visual analytics techniques. The system will feature interactive charts and graphs, enabling users to examine match data in depth using bar charts, line graphs, and heatmaps. Additionally, it will include filtering and querying options so that users can focus on specific teams, players, venues, or match formats. The dashboard will also facilitate comparative analysis, helping users draw meaningful insights by comparing player statistics, team performances, and match conditions across different matches and tournaments. Finally, the system will be designed to effectively present data-driven stories, offering a compelling narrative on cricket trends, significant performances, and key factors influencing match outcomes.

6 Tech Stack

- **Backend:** Python (Flask/Django) and Java, Pandas for data processing
- **Frontend:** JavaScript (React.js or Vue.js), D3.js/Plotly for interactive visualizations
- **Database:** PostgreSQL/MySQL for storing processed data

7 Team Members & Responsibilities

Each member will contribute to different aspects of the project:

- **Data Processing & Cleaning & Analytics:** Mridul Pandey (210634), Rudransh Goel (210880)
- **Backend Development:** Md Rahbar (210601), Shreyas Kumar (218071009)
- **Visualization Development:** Goural Dureja (210393), Dhruv Misra (210341), Pratyush Amrit (210762)
- **Frontend Development:** Aayush Aggarwal (210019), Peeyushi Yadav (210717)

This project aims to leverage big data analytics and visualization techniques to enhance the understanding of cricket statistics. Our system will serve as an engaging platform for cricket fans, analysts, and professionals to explore historical cricket data interactively.